



United International University
Department of Computer Science and Engineering
Course Code: CSE 2215 Course Title: Data Structure and Algorithms I (B)

Class Test – 3
Duration: 30mins (all inclusive)

Date: 17/08/2021
Marks:20

1. Convert the following infix expression into postfix expression using a stack. Use the infix expression after your student id in the following table. Show every step of your work and draw a stack for each step. [20]

Student ID	Infix expression
011173044	$((8*(4+9))*((8+7)*(3*9)))$
011181265	$((7*7)*(4-7))+((4*9)-(6+3))$
011183017	$((1-6)*7)-((9+2)-(5*4))$
011183020	$((4*2)-(4-3))+((1+8)+(8+8))$
011183056	$((2+3)*7)-((2-1)*(3+3))$
011183077	$((6*9)+(4-5))+((6-8)+4))$
011191160	$((4*4)*(3-8))*((4+4)-(9-7))$
011191173	$((5+3)+(4-5))-((4+3)*(5+8))$
011191179	$((9-4)*(6-7))*((6+5)+(2*9))$
011191212	$((3-3)+(3*7))-(2+(5+5))$
011192083	$((3*5)-(5*4))+((2-9)*(3-6))$
011192094	$((8+1)-(2+8))-((9+1)*(9*4))$
011192122	$((6+(2-1))-((5*8)*(4-8)))-8$
011193028	$((1*6)+5)+((1*3)+(4-6))$
011193038	$((6*6)*(1+3))-((9-9)*(3*6))$
011193040	$((9+1)-(4*7))-((8+1)*(4*9))$
011193045	$((9+7)+(1*4))-((5+3)+(1-1))$
011193047	$((8-3)-(8+8))-((7-8)*(4+8))$
011193052	$((1*7)*1)*((4-9)-(7-6))$
011193068	$((3+4)+(7-8))*(5-(5*8))$
011193122	$((5+3)-5)*((2+3)*(4+8))$
011193123	$((2*2)-9)+((2-7)*(4*5))$
011201019	$((7+2)+(9+3))*((2-7)*(7*6))*9$

011201031	$((9+1)-(6+7))-((9+9)+(1*1))$
011201037	$((7*3)+(3+1))*((1-8)-(5+4))$
011201084	$9-(((5+2)-(9*7))+((4-2)+(8*3))))$
011201088	$8*(((6-5)*(8+2))-((4-7)*(2*3))))$
011201108	$((4+3)+(1*1))+((9-4)*3)$
011201125	$3*(((7+1)-8)*((5-9)+(3*9))))$
011201128	$4+(7-((4*7)+(2*5))))$
011201156	$((7+1)-(1+5))+((3+3)-(8+5))$
011201166	$((2-2)+(6*5))*((7*1)-(5+3))$
011201170	$3-(((1-7)-(2*1))+((1*8)-(5-1))))$
011201173	$((8-6)*(6-3))+((5*7)+(4*4))$
011201176	$((8*6)+(7-3))+(3*(2-6))$
011201286	$((7+2)*1)-((3*5)+(7*9))$
011201334	$((8+8)+(4-8))-((8*6)+(9*1))$
011201344	$((9*7)*(1+3))*((8+8)-(3*5))-7$
011201352	$((5*7)-(1-6))*((3+8)-(9*5))$
011201353	$((9+8)*(8+7))*((8+7)*7)$
011201366	$((1+(6*4))+((6+4)+(2-5))$
011201400	$((7+5)*(2-2))*((6-8)*(9-7))$
011201403	$((7*8)*6)*(4+(3+9))$
011201406	$4-(((3*1)*(2-2))-((2-1)*2))$
011201446	$4-(((6*3)+(7*7))-((9-6)+(8+3))))$
011202199	$((7*6)*5)*((8+5)-(1*3))$
Not registered	$((7-8)-(5-6))-((7*1)*(1+2))$